Evidence Appendix

REL consolidated technical report v0.7

5 November 2026 | Historical agent studies and separate component evidence

A. Study register

ID	Date and setting	Executed scope	Principal limitation
S1	18 Oct; JSON Schema transfer	Two origins and 24 later native sessions; separate direct replays	Unchanged origin code already sufficient; two origin histories
S2	18 Oct; changed semantics	24 adaptations; 24 later unchanged-code replays; separate scope diagnostic	One source family; influential block; perfect score missed a semantic defect
S3	24 Oct; connected migration	13 native task sessions: acquisition plus four three-episode continuations	One constructed sequence; every arm retained executable code
S4	24 Oct; costly observation rediscovery	26 native task sessions: two acquisitions plus four six-task policies	Two generated streams; cheap actual database runtime; whole-system cost unknown
S5	24 Oct; raw versus compact observations	Six acquisitions plus 24 later sessions	Six paired cases; all outcomes correct; cache allocation uncontrolled
S6	24 Oct; restored result values	Six acquisitions plus 24 later sessions	Six new paired cases; candidate failed one declared cost criterion

S1 and S2 share origin assets. S3/S4 have policy-generated later histories. S5/S6 reuse a task family but have different fresh constructed cases. Summing rows across these studies would not yield an independent sample size. S2's 24 replays are executions, not 24 new model sessions.

The accompanying `evidence/` directory retains the current dated public-safe study summaries, including intermediate 19-23 October work. `reproduce-study1/` and `reproduce-study2/` preserve sanitized attempt tables, source digests and offline accounting scripts. Their source records remain outside this bundle; arithmetic reproduction does not authenticate native events.

B. Outcome and resource definitions

Task completion means the saved artifact met its declared scored contract. It is not complete specification correctness. First saved decisions, final decisions, tool errors, corrective actions, source validity and lesson invocation are separate observations. No missing field is interpreted as zero.

Total tokens = input + output. Uncached input plus output = total tokens - cached input. Reasoning-output counters are already included in output. Cached input is not added to total again. Token quantities are not currency, hardware energy or total research expenditure.

For S4-S6, query work includes diagnostic probes and final execution. SQLite VM counts are conservative instruction upper bounds; the database work is not a direct measure of remote model inference. CPU and wall timing are host-specific and background work is not fully controlled.

A deployment comparison may charge the full common acquisition to each alternative. That is counterfactual allocation; the actual campaign counts common acquisition once. Costs of parent-model planning, all engineering, user time and physical/remote energy remain unmeasured. No positive energy conclusion is available.

C. Selected arithmetic checks

Quantity	Reconstruction	Rounded result
S1 inclusive REL versus equal-history ordinary	1,473,718 / 1,475,038 - 1	-0.09%
S4 later REL facts versus ordinary facts, tokens	1,108,706 / 713,384 - 1	+55.42%
S4 later REL facts versus no history, query work	1 - 1,136,900 / 4,909,400	76.84% reduction
S4 later ordinary facts versus no history, query work	1 - 1,101,500 / 4,909,400	77.56% reduction
S6 charged restored REL versus compact REL, tokens	(805,136 + 634,348) / (1,021,888 + 634,348) - 1	-13.09%
S6 charged restored REL versus compact REL, uncached + output	(146,704 + 141,420) / (128,192 + 141,420) - 1	+6.87%
S6 charged ordinary versus restored REL, tokens	(564,115 + 634,348) / (805,136 + 634,348) - 1	-16.74%

The included `verify_selected_results.py` loads the supplied original summaries, reconstructs these comparisons and checks declared totals against their parts. It also invokes the existing sanitized S1/S2 arithmetic scripts. It does not infer a causal effect, perform a power calculation or rerun model tasks.

D. S6 complete later-policy summary

Policy	Tasks	Total tokens	Uncached + output	Query VM work
R: raw REL	6/6	871,285	153,461	1,302,200
C: prior compact REL	6/6	1,021,888	128,192	1,302,200
V: restored-results REL	6/6	805,136	146,704	713,400
H: ordinary restored results	6/6	564,115	128,275	1,290,600

Common acquisition: 634,348 tokens, 141,420 uncached input plus output, and 4,256,100 query-work units. Actual study usage: 3,896,772 total tokens, including 3,198,720 cached input tokens. Acquisition and the four policy totals reconcile exactly; allocation is not double-counted.

S6 prespecified selection check	Result
At least four paired total-token reductions	Pass
Charged total tokens at least 10% below C	Pass
Charged total tokens not above R	Pass
No extra failed decisions	Pass
Query work within 10% allowance	Pass
Uncached input plus output not increased	Fail

The failing check is preserved. It is a scoped engineering selection rule, not a law that every resource must improve simultaneously. Changing that rule after observing the result could inform future deployment preferences but would not make this candidate pass its original study.

E. Real-use observations: bounded examples

OBS1, packaged response behavior. A saved preview response had baseline policy headers; the same recorded 404 path in the packaged handler lacked them. The corrected receipt preserved status 404 and included the headers. This supports checking the actual consumer path. The observer did not perform a new live security audit, and the correction is not randomized evidence of REL's effect.

OBS2, decision-to-action enforcement. Native events recorded a memory-admission block followed by 51 completed tests. A later lighter validation reported admission and 65 passing tests, with one intentionally excluded. The project acknowledged the original orchestration error and coupled admission to execution. A successful admitted path alone does not prove that every rejected path now prevents work.

These events occurred on 23 October UTC and were reviewed on 24 October local time. Private project names, account details, transcript text and paths are omitted. The observer's source bindings remain private. No substantive project changes or shared-skill edits were made by the observer.

F. Original brain-AI strand

The broad programme asked how cheap prediction, working state, observation memory and slower updates might improve adaptive systems. Run68, completed 13 October, tested an exposed-input intervention on the Run67 forecasting source. Removing the standardized cap on the current-value feature improved the primary native normalized MSE by 25.834%, across 17,338 scored labels, against the same clipped predecessor. Peak normalized MSE improved by 47.656%.

The primary candidate still had 0.550% worse aggregate MSE than the simple latest-observation predictor, and local/tail harms remained. The intervention changed the fitted and online procedure together; it was not an inference-only experiment. It reused already-exposed source inputs and was not independent confirmation. No claim of biological equivalence, a new architecture, measured energy superiority or general RSI follows.

This is historical context. The programme subsequently tested acquired methods, output-directed computation and source transfer. Those results are registered below; the old uncapped-history proposal is no longer the next action. No brain-AI result is added to agent success denominators or attributed to the current REL wording.

G. Current component register: separate evidence family

The Run120 component map identifies nine historical components, not nine independent architecture replications. It binds 45 named historical source/method artifacts and records compatible and incompatible interfaces. Detailed reports remain accessible through the private repository baseline; selected numerical summaries and the runnable reference are supplied here.

ID	Run	Retained capability	Scope or counterevidence
C1	85	Guarded SQL methods and context reuse; 32 correct episodes	Joint guidance used 32.96% more tokens than ordinary
C2	102	Learner-authored primal-dual output checker	Conventional mathematics; finite floating-point allowance, not formal rounding proof
C3	103	Solver continuation and output-directed precision	Saved vectors do not preserve solver factorization

ID	Run	Retained capability	Scope or counterevidence
C4	106	Two acquired update programs transferred to later recordings	Lower MSE against later robust comparator; 16 versus 11 above-band outputs
C5	111	Input acquisition combined with state and checker	Ordinary policy provides request savings; learned twin doubles wall time
C6	112	Incremental aggregate maintenance and restart	Ordinary implementation covers the measured efficient frontier
C7	113	Temporal streaming/frontier/index and recovery	10,240 agreeing decisions; index was 2.80 times streaming on conference workload
C8	115	Ordinary checked graph state	5,760 correct deliveries; extra certificate packaging missed benefit screen
C9	119	Two-history equal averaging and diagnostic readouts	Exposed-source trade-off; no readout passed the full joint criterion

Run118 is a separate direct instruction pilot: both arms used the same history; REL's extra core instructions were the treatment. Run106's ordinary and joint-guidance learners both used REL. Consequently a Run106 gain cannot establish REL's incremental effect. Run120's source-free reference tests and Run121's source qualification do not increase model-session or physical-recording sample size.

H. Localization reconstruction and transfer admission

The supplied `component-evidence/RESULTS119.json` contains previously saved trajectory-level metrics and summaries, not raw sensor/pose arrays. `verify_component_results.py` reconstructs the six-recording table, equal recording weighting, two-replica averaging, percentage changes and failed qualification flags. The six recordings are 4 through 9. Counts are repeated, correlated endpoints; their replica means may be half-integers. This verifies arithmetic, not independent data collection or upstream model execution.

The Run120 reference under `reference/` runs without credentials or raw data. With Python and the pinned NumPy dependency, `python -B example.py` and `python -B test_reference.py` exercise its constructed case and 17 engineering checks. Dependency installation is not performed automatically. Refer to its README for the trusted fixed-record scope and tested versions. Full original study hosts and all historical implementations are not bundled.

Run121's `source-review/SOURCE_REVIEW.md` and JSON record author-page/paper coverage and inaccessible README/configuration. The derived range-only design in METHODS_RESEARCH.md is prospective. No raw AAU arrays, new filter scores, new training or native experimental learner sessions were produced. Required geometry, clocks, reference uncertainty/coverage and calibration/evaluation roles remain unresolved; the source-admission verdict is NOT_ADMITTED.

The six-recording mean gains reconstruct as `100*(1-0.03774123/0.04211678)` for balanced MSE and `100*(1-832.5/920)` for above-band counts. The cost against acquired MSE is `100*(0.03774123/0.03705959-1)`. The verifier uses full-precision saved values. Neither these identities nor the Cartesian averaging identity supplies a tail-risk guarantee.

I. Versions, preservation and sharing

The historical v0.5 private repository was aligned at commit `507357e1d7a55335f9205134ad1c32ac4657b0bc`. All 20 installed core files matched its actual Git blobs and the local core manifest immediately before alignment; the alignment changed only 13 documentation/evidence files. Installed SKILL.md SHA-256: `f598cee5206dd415d5884f557442ec29d07dd4049bbcde23d9c9bda214f16c1b`.

Historical study policies are identified by their own evidence, not retroactively by that digest. The old v0.3/v0.4 manuscripts remain preserved. The preserved v0.5 PDF/ZIP and manuscript-before-v0.6 retain that edition.

Version0.7 is now the reader entry; v0.6 is also preserved unchanged. S1-S6 result tables are unchanged. The v0.7 sharing review uses source baseline af68dc3b3b6f34157f0e7718d5f3e326856c3c64. The source baseline for the new component material is private commit `e775ee8b68a2db210641768024e9bdae3384dac0`. The package records the current installed SKILL.md identity separately from historical treatments; final delivery commit is recorded outside its self-hashed package.

Package manifests bind supplied bytes and relative paths. The author authorized this public reader export on 5 November 2026; the broader research repository remains private. Public availability does not imply peer review, an open-source license grant or access to the full historical archive. A full scientific replication would additionally require a qualified runnable study host and complete releasable task corpus.

J. Whole-programme lineage and restored counterevidence

This is a retrospective synthesis, not a new audit of every raw record from every run. The v0.7 review checked S1-S6 against original saved usage and outcome records, inspected the governing programme and component register across the broader research arc, and traced selected later claims to their original reports and summaries. Historical literature reviews are inherited at their recorded scope. No original learner trial or physical recording was rerun for this revision.

Strand or study	Contribution to this report	Evidence boundary
Public-agent traces	Source identity and annotation/reconstruction discipline	Separate investigation; no new incident established or incident corpus merged here
Biology and early brain-AI work	Cheap prediction, working state and slower updates as questions	No biological experiment or measured energy equivalence; early model-internal interventions are outside this report's fixed-weight study claims
OpenAI mathematics and Alien review	Conditional constructions, alternative representations and mechanism comparisons	Dated source reviews; not independent proof replication or borrowed novelty
Run86, longer lifecycle	64 correct reports, 60 later direct executions, four chains/two public-data families	Ordinary and REL both reuse; overlapping windows; REL retains 26,475 extra acquisition tokens
Run92, procedure revision	One authored revision, eight fresh learners; identical selected methods/later outputs	Evolved procedure adds 20.12% tokens including revision; candidate retired
Run106, retained scalar	Second joint learner: 2.42% lower MSE and 14 vs 22 above-band outputs versus scalar reset	Same physical/host state remains; first learner removed its scalar; two later recordings
Run111, ordinary schedule alternative	Alternate/progress buys 564 fewer inputs than witness/progress; witness is 8.77% faster	Does not isolate unique witness or learned-method superiority
Run114, selection and later counterexample	Twelve chooser/writer cells; ordinary, diversity and compatibility procedures	Compatibility candidate fails; later row-order defect separately repaired, original outcome retained

The added findings are in `component-evidence/REPORT86.md`, `REPORT92.md` and `REPORT114.md`; `ADDED_EXPORTS.json` records their source/delivered digests and transformations, while `EXPORT_LINKS.json` retains the earlier source mappings. These source snapshots document original findings; their then-current next actions are historical, not this report's current plan.

Computational lineage	Source and transformation	What is evaluated
Symbolic/reporting	UCI Appliances energy prediction and Beijing PM2.5 observations; controlled SQL/contract changes	Correct reports and executable reuse, not actual instrument calibration

Computational lineage	Source and transformation	What is evaluated
Numerical	LIBSVM w8a, a9a and mushrooms entries; OSQP-derived objectives and explicit curvature assumptions	Optimization-output bounds and operation costs, not dataset-label predictive accuracy
Temporal graph	SocioPatterns workplace [11], conference and SFHH contacts; controlled event/query changes	Declared temporal reachability semantics, not a social/medical outcome
Localization	UTIAS MR.CLAM [8] motion, range/bearing and reference streams	Recorded replay error under supplied map/initialization roles, not deployed navigation safety
Independent candidate	AAU industrial range-only AMR author source [6]	Source-contract review only; not admitted for scoring

References [6]-[13] are in PAPER.md. Exact source roles, changes and reading scopes remain in the supplied component map and source review. No raw UCI, LIBSVM, contact or localization records are distributed here.

K. Ordinary nonmodel controls and reproduction coverage

S4 included researcher-implemented nonmodel controls:

Policy	Later tasks	Later query VM work	Acquisition VM work
auto	0/6	2,227,100	700,400
measure_all	6/6	6,922,600	2,139,100
revision_cache	6/6	1,404,400	2,139,100

The revision cache keys source revision and filter parameters. Construction effort is unmeasured; these values establish neither a general optimizer guarantee nor agent superiority. Source: the bundled S4 evidence summary, derived from its original NONMODEL_BASELINES.json. Model-token comparisons do not include an invented token charge for these deterministic controls.

The package offers three distinct checks: saved-summary arithmetic for S1-S6; trajectory-summary arithmetic for Run119; and a constructed localization runner with 17 engineering tests. A new extraction tests file completeness and executable delivery. It is neither a new independent dataset nor a security sandbox, and it cannot establish that the upstream measurement or author is correct.

`rel-current/` supplies the 21 files in the current generic-core manifest, including SKILL.md and its supporting references/helpers. It is explanatory reader context, not a historical treatment reconstruction. Exact historical full prompts, native hosts, transcripts and physical arrays are not included. Their absence prevents full original-trial replication from this ZIP. The present manuscript review and targeted privacy scan do not certify arbitrary future file additions.